EFY TESTED
AKSHITA GOEL & SANI THEO

Arduino-Based COIL WINDING TEMPERATURE RECORDER And ALARM GENERATOR

NAVPREET SINGH TUNG

Monitoring the temperature of a transformer or electric motor winding involves data acquisition. This project is developed to monitor the temperature of an electric motor and a transformer with high accuracy, incorporating display and alarm facility. This circuit can be installed near a transformer winding or a motor's location, or it can be attached to either of these using suitable support. It can track the temperature of the device or equipment and its surroundings.

The project includes LM35 temperature sensor to detect the temperature in the device (transformer/motor), buzzer to alert abnormal condition in the device and LCD (LCD1) to display the current status of the device. The author's prototype is shown in Fig. 1.

Transformers and electric motors are the most significant sources of electrical power and industrial systems. Failure of these devices could cause electrical power outages, environmental health hazards and more. Failure can happen due to various reasons, including dielectric failure, winding deformation, short-circuit, magnetic circuit, electrical disturbance, distortion of insulation, lightning, insufficient maintenance, slack connection and overloading.

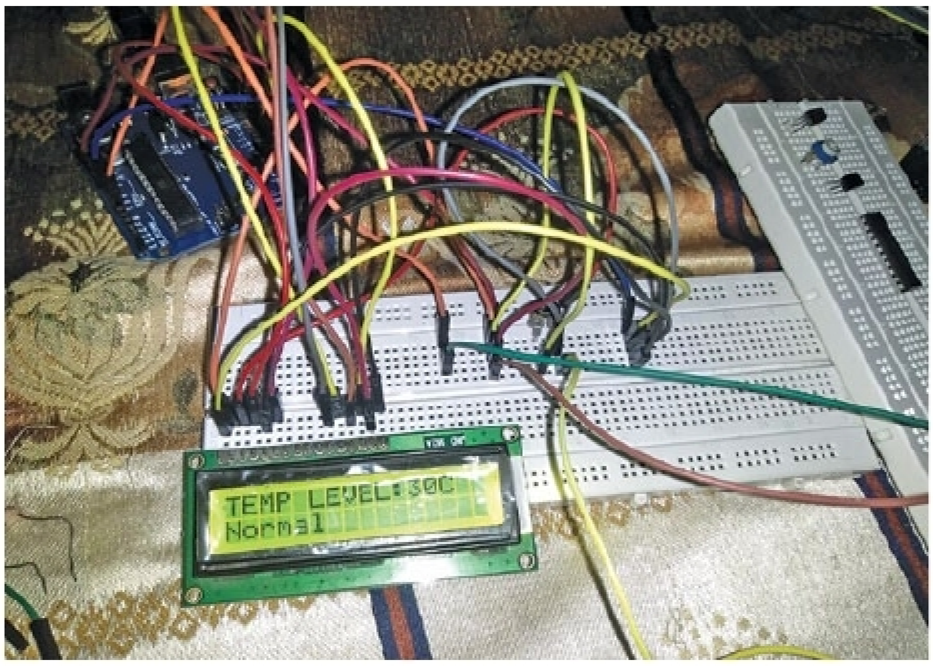

Fig. 1: Author's prototype

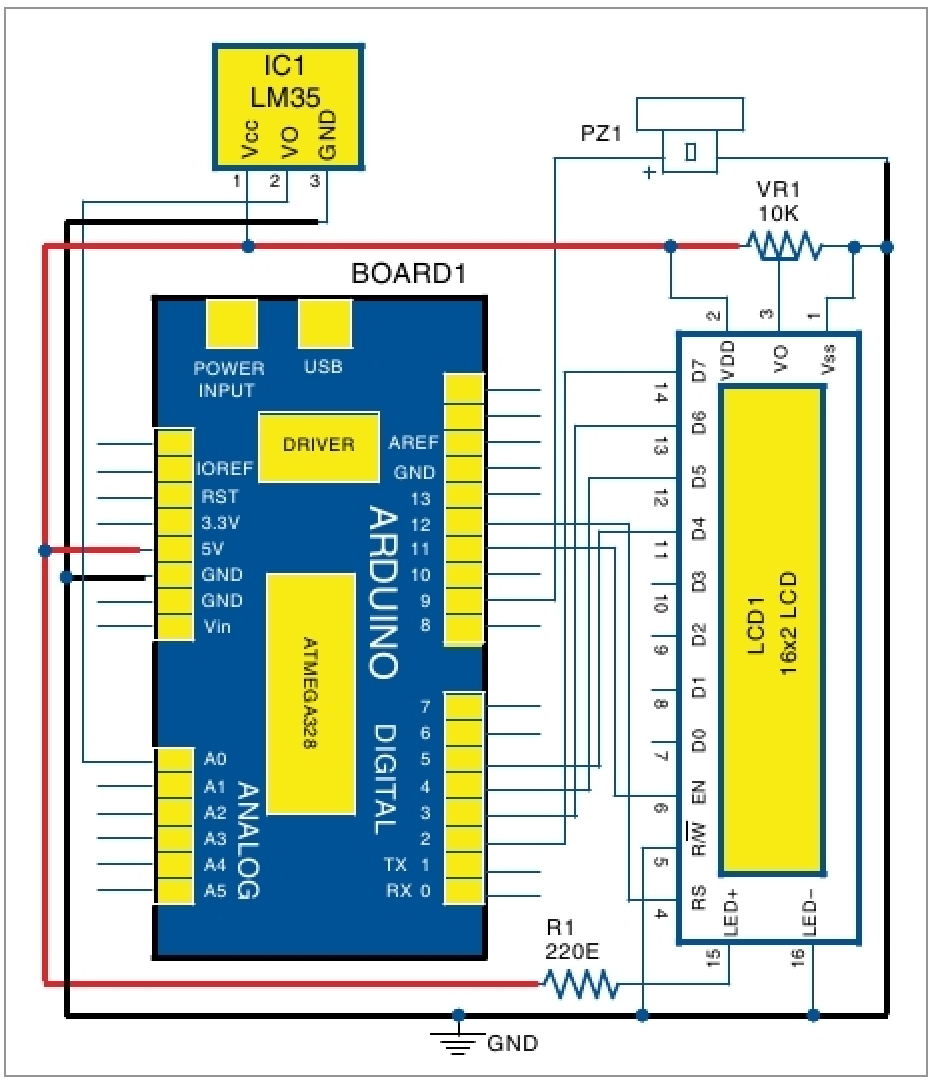

Fig. 2: Circuit diagram of coil winding temperature recorder and alarm generator

Functional life of these devices is evaluated by their capability to dissipate the generated heat to the surrounding environment. Evaluation of temperatures can provide an insightful diagnosis of the devices.

The penalty for temperature rise may not be sudden, but gradual, as long as it is within break-down limit. Among these consequences, insulation degradation is important. Since insulation is costly, its deterioration is unwanted. Determination of oil temperature and winding insulation system temperature is critical.

Circuit and working

The circuit diagram of the coil winding temperature recorder and alarm generator is shown in Fig. 2. It comprises LCD1 (16x2), Arduino Uno, piezo buzzer (PZ1), resistor (220-ohm), preset (10-kilo-ohm) and 9V battery.

LM35. It provides linear output proportional to temperature, with 0V corresponding to 0°C and output voltage change of 10mV for each °C change. It is easier to use than a thermistor and a thermocouple.

Output of LM35 can be connected directly to Arduino analogue input. Since Arduino analogue-to-digital converter (ADC) has a resolution of 1024 bits, and reference voltage is 5V, the